MA385: Assignment ANS with outline solutions (Due **5pm, Friday 23 October**)

This assignment contributes 10% to your final grade for MA385. Your solutions must be clearly written (ideally, by hand), and neatly presented: up to 20% of the marks will be given for clarity and correctness of exposition.

For numerical calculations (such as in Q5(b)), you may make use a calculator or computer; details of the calculations are not required.

Write your name and ID number on the first page. Number every page. Upload a scanned PDF version to canvas at https://universityofgalway.instructure.com/courses/58642/assignments/165065.

If you really want, you can submit a hard copy at the lecture on the 22nd of October. If doing so, staple the pages together, and write your name and ID number at the top of each *page.*

The assignment is collaborative*: you are allowed work with classmates. However, each of you needs to submit your own hand-written version of your solutions, and you should include a clear statement on who you collaborated with, and on what aspects.*

Q1. (Based on Q3 from Tutorial Sheet 1) Show that the Secant iteration can be written as

$$x_{k+1} = \frac{x_k f(x_{k-1}) - x_{k-1} f(x_k)}{\delta_k},$$

for $k = 1, 2, \ldots$, and give a formula for δ_k.

Answer: The method is usually written as

$$x_{k+1} = x_k - f(x_k)\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}.$$

Clearly the right-hand side is the same as

$$x_k\left(1 - \frac{f(x_k)}{f(x_k) - f(x_{k-1})}\right) + x_{k-1}\frac{f(x_k)}{f(x_k) - f(x_{k-1})},$$

which simplifies as

$$x_k\left(\frac{-f(x_{k-1})}{f(x_k) - f(x_{k-1})}\right) + x_{k-1}\frac{f(x_k)}{f(x_k) - f(x_{k-1})}.$$

This gives the desired expression with

$$\delta_k = f(x_{k-1}) - f(x_k).$$

Q2. In Q5 of Tutorial Sheet 2, you deduced that, if the Secant Method computes the iterates $\{x_0, x_1, x_2, \ldots\}$ then

$$\tau - x_{k+1} = -(\tau - x_k)(\tau - x_{k-1})\frac{f[x_{k-1}, x_k, \tau]}{f[x_{k-1}, x_k]}, \quad (1)$$

where

$$f[a, b] := \frac{f(b) - f(a)}{b - a}$$

and

$$f[a, b, c] := \frac{f[b, c] - f[a, b]}{c - a}.$$

(a) Use Equation (1) to give a justification (does not have to be rigorous) that there is a constant K such that, as $k \to \infty$, we get

$$|\tau - x_{k+1}| \approx K|\tau - x_k||\tau - x_{k-1}|.$$

(b) Use this to deduce that the Secant Method converges with at least order $q = (1 + \sqrt{5})/2$.

Answer: (a): We assume the method converges. Therefore $x_k \to \tau$. Let $\delta_k := x_k - x_{k-1}$. Then

$$f(x_{k-1}, x_k) = \frac{f(x_{k-1} + \delta_k) - f(x_k)}{\delta_k}.$$

Since, as $k \to \infty$ we get $\delta_k \to 0$, then $f(x_{k-1}, x_k) \to f'(\tau)$. Repeating this for $f(x_{k-1}, x_k, \tau)$, you should get $f(x_{k-1}, x_k, \tau) \to Cf''(\tau)$. (Actually, that $C = 1/2$, but it is not important). This now gives is that $\frac{f[x_{k-1}, x_k, \tau]}{f[x_{k-1}, x_k]} \to K$, where K is a constant (since $f'(\tau)$ and $f''(\tau)$ are constants).

Answer: Part (b): Suppose that the order of convergence is q, so that $\lim_{\infty} \frac{|\tau - x_{k+1}|}{|\tau - x_k|^q} = \mu$ for some constant μ. Let's infer (not rigorously) that

$$|\tau - x_{k+1}| \sim \mu|\tau - x_{k-1}|^q. \quad (\star)$$

We'll use this twice.
First, this gives that $|\tau - x_{k+1}| \sim \mu^2|\tau - x_k|^{q^2}$.
Next, from Part (a), one can infer (again, non-rigorous) that

$$|\tau - x_{k+1}| \sim K|\tau - x_k||\tau - x_{k-1}|.$$

Again use $(\star)$ to get $|\tau - x_{k+1}| \sim \mu K|\tau - x_{k-1}|^{q+1}$.
Matching the powers of q in the two expressions involving $|\tau - x_{k+1}|$ and $|\tau - x_{k-1}|$ we get that $q^2 = q + 1$.
The only positive solution to this is $q = (1 + \sqrt{5})/2$.

Q3. Suppose that we have a fixed point iteration (FPI) method $x_{k+1} = g(x_k)$ which we know to be converges to fixed point of g, denoted τ. Show that the method converges with at least order p if

$$g'(\tau) = g''(\tau) = \cdots = g^{(p-1)} = 0.$$

Answer: A method converges with at least order p if there is a constant $\mu \geqslant 0$ such that $\lim_{k\to\infty} \frac{|\tau - x_{k+1}|}{|\tau - x_k|^p} = \mu$. So, we need to find μ such that $\frac{|\tau - x_{k+1}|}{|\tau - x_k|^p} \to \mu$. Let's write out a Taylor series for $g(x)$ about τ:

$$g(x) = g(\tau) + (x-\tau)g'(\tau) + \frac{1}{2}(x-\tau)^2 g''(\tau) + \cdots + \frac{(x-\tau)^{p-1}}{(p-1)!} g^{(p-1)}(\tau) + \frac{(x-\tau)^p}{p!} g^{(p)}(\eta)$$

for some $\eta \in [x, \tau]$. Since $g'(\tau) = g''(\tau) = \cdots = g^{(p-1)} = 0$, this simplifies to

$$g(x) = g(\tau) + \frac{1}{p!}(x-\tau)^p g^{(p)}(\eta).$$

Now take $x = x_k$, so this becomes

$$\underbrace{g(x_k)}_{x_{k+1}} - \underbrace{g(\tau)}_{\tau} = \frac{(x_k - \tau)^p}{p!} g^{(p)}(\eta).$$

for some $\eta \in [x_k, \tau]$. Since $x_{i+1} = g(x_k)$, and $g(\tau) = \tau$, we now have

$$|x_{k+1} - \tau| = \frac{|x_k - \tau|^p}{p!} |g^{(p)}(\eta)|.$$

Rearranging:

$$\frac{|x_{k+1} - \tau|}{|x_k - \tau|^p} = \frac{|g^{(p)}(\eta)|}{p!}.$$

To finish, since we know the method converges, $x_k \to \tau$. So, since $\eta \in [x_k, \tau]$, it must be that $\eta \to \tau$, as $k \to \infty$. Thus $g^{(p)}(\eta_k) \to g^{(p)}(\tau)$, which is a constant.

$$\frac{|\tau - x_{k+1}|}{|\tau - x_k|^p} \to \mu \qquad \text{where } \mu = \frac{|g^{(p)}(\tau)|}{p!}.$$

Q4. Suppose we want to estimate $\sqrt{3}$, via FPI using a method of the form

$$x_{k+1} = a x_k + b/x_k, \qquad \text{for } k = 0, 1, 2, \ldots. \quad (2)$$

Show that one needs to choose $b = 3 - 3a$.

Answer: What we need to do is show that $g(\sqrt{3}) = \sqrt{3}$ where $g(x) = ax - b/x$. In this case $g(x) = ax + (3-3a)/x$. Therefore,

$$g(\sqrt{3}) = a\sqrt{3} + \frac{3-3a}{\sqrt{3}} = a\sqrt{3} + \frac{3}{\sqrt{3}} - \frac{3a}{\sqrt{3}} = a\sqrt{3} + \sqrt{3} - \sqrt{3}a = \sqrt{3},$$

as required.

Q5. Take $a = 3/4$ in Equation (2).

(a) Show that the resulting g is a contraction on $[1, 3]$.

Answer: First we need that $1 \leqslant g(x) \leqslant 3$ for all $x \in [1,3]$. Note that $g(1) = 3/2 > 1$ and $g(3) = 5/2 < 3$. Also, since $g'(x) = 3(1 - 1/x^2)/4 > 0$, this is a monotonically increasing function. So $1 \leqslant g(x) \leqslant 3$ for all $x \in [1,3]$. Next, note that $1 - 1/x^2 \in (0,1)$ so $|g'(x)| < 1$. Therefore it is a contraction.

(b) Take $x_0 = 1.5$, and compute the corresponding values of x_1, x_2 and x_3. Compute the errors $\mathcal{E}_k = |\tau - x_k|$ for $k = 0, 1, 2, 3$.

Answer:

$$x_1 = 1.625, \quad x_2 = 1.68029, \quad x_3 = 1.70657.$$

$$\mathcal{E}_0 = 0.23205, \quad \mathcal{E}_1 = 0.107051,$$
$$\mathcal{E}_2 = 0.051762, \quad \mathcal{E}_3 = 0.025483.$$

(c) Show that $\mathcal{E}_{k+1}/\mathcal{E}_k$ is roughly constant. What can we infer from that?

Answer: $\mathcal{E}_1/\mathcal{E}_0 = 0.4613249$, $\mathcal{E}_2/\mathcal{E}_1 = 0.4835306$, $\mathcal{E}_3/\mathcal{E}_2 = 0.4922986$. This implies that there is a constant $\mu \in (0,1)$ such that $\mathcal{E}_{k+1}/\mathcal{E}_k \to \mu$. That is: it appears the methods converges almost linearly (i.e., order 1).

Q6. Determine the values of a and b in Eq (2) that would correspond to Newton's method applied to solving $x^2 - 3 = 0$. Show that this method converges with at least order 2.

Answer: Taking $f(x) = x^2 - 3$, and $g(x) = x - f(x)/f'(x)$, with a bit of rearrangement we get $g(x) = x/2 + 3/(2x)$. That is $a = 1/2$ (and so $b = 3/2$).

There are a few approaches to showing this converges with at least order 2 (we'll assume we know it converges at all). These include checking that the conditions required by Theorem 1.5.2 are satisfied. But a little better wound be to use the result in Q3 with $p = 2$. That is, show that $g'(\tau) = 0$. Here $g'(x) = (1 - 3/x^2)/2$. So $g'(\sqrt{3}) = (1 - 3/3)/2 = 0/2 = 0$.